

# Examining Multilingual Embedding Models Cross-Lingually Through LLM-Generated Adversarial Examples

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## Creating Cross-Lingual Semantic Search Evaluations for Specific Domains and Language Pairs

### Scenario

How to choose the right embedding model for semantic search across languages?

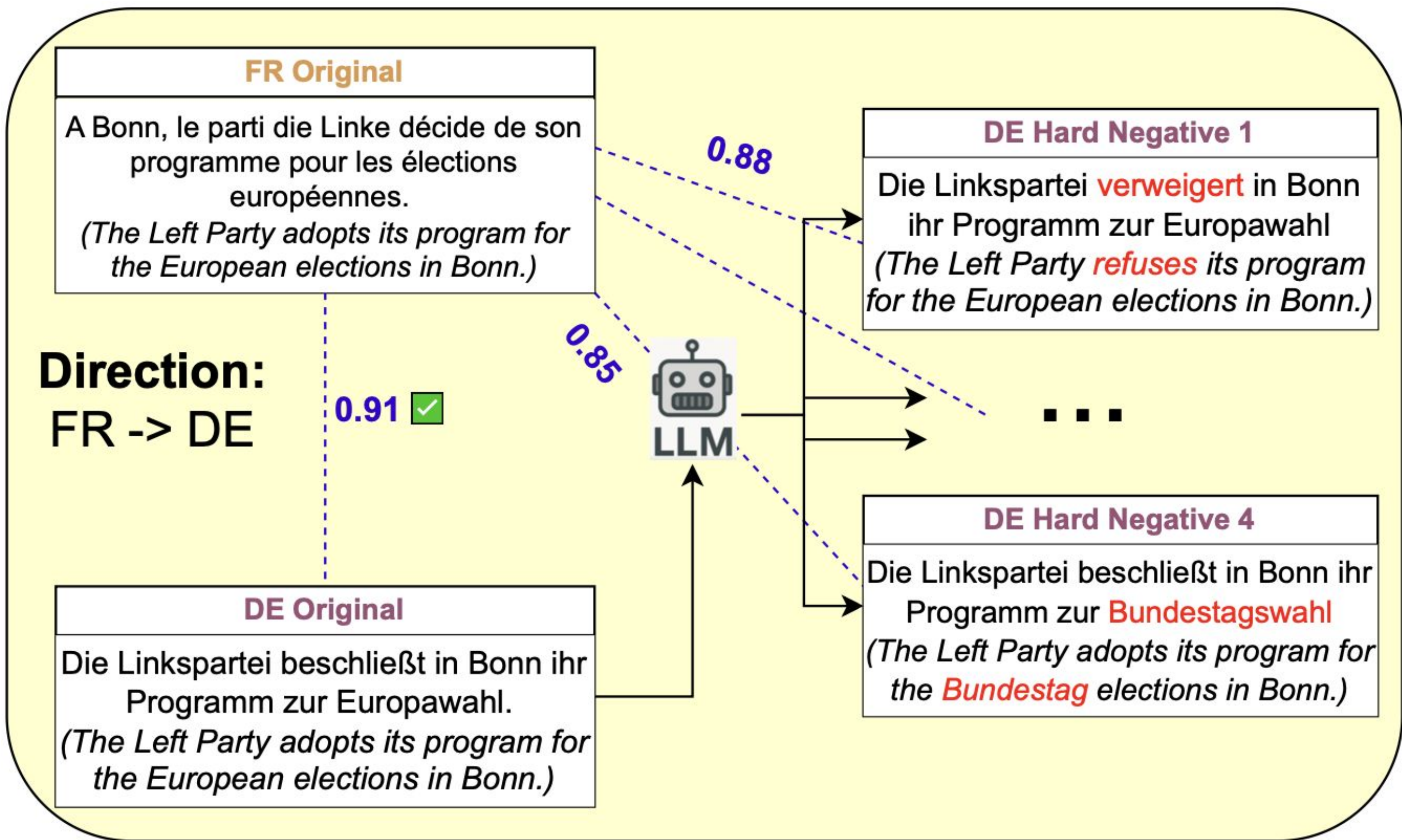
### Current State

**Broad coverage, limited focus:** Benchmarks cover many languages and tasks, but few directly test cross-lingual semantic retrieval ( $X \leftrightarrow Y$ ).  
**English-anchored evaluation:** Tasks like X-STS and Bitext Mining test cross-lingual retrieval, but mainly via English pairs (EN- $X$ , EN- $Y$ ).  
**Questionable generalization:** English-based results often fail to reflect performance on direct  $X \leftrightarrow Y$  retrieval.

### Our Proposal

1. Use **parallel texts** from the target domain.
2. Make **LLM-generated hard negatives**: look similar, mean different.
3. Test whether models **rank the true translation top**.

### Examples and Observations



- Prompt asks to keep the syntactic form, but change the meaning.
- Named entities, verbs and nouns are often modified.
- Polarity of the texts are often inverted.
- One- or two-word replacements are the most effective distractors.

**Manual Checks:** Over 98% of the distractors meet the intended “Similar-form, different meaning” criterion

### Cross-Lingual Semantic Discrimination (CLSD) Task

$\text{sim}(X\text{-original}, Y\text{-original}) \gg \text{sim}(X\text{-original}, Y\text{-hard negative})$

**Precision@1** = How often the model ranks the original translation  $Y$  above hard negatives.

### Results

| Model (Hugging Face Name)                                                | Alignment     |             |            | WMT19        |              | WMT21        |              | Average      |
|--------------------------------------------------------------------------|---------------|-------------|------------|--------------|--------------|--------------|--------------|--------------|
|                                                                          | Cross-lingual | Para-phrase | Retrie-val | DE→FR        | FR→DE        | DE→FR        | FR→DE        |              |
| Cross-Lingual CLSD Evaluation                                            |               |             |            |              |              |              |              |              |
| multilingual-e5-base                                                     | ✓             | ✓           | ✓          | 91.51        | 88.46        | 86.34        | 81.97        | 87.07        |
| multilingual-e5-large                                                    | ✓             | ✓           | ✓          | 94.50        | 91.51        | 91.38        | 87.57        | 91.24        |
| gte-multilingual-base                                                    | ✓             | ✓           | ✓          | 90.22        | 89.68        | 90.48        | 91.60        | 90.50        |
| paraphrase-multilingual-mpnet-base                                       | ✓             | ✓           | ✗          | 91.31        | 91.11        | 91.15        | 92.95        | 91.63        |
| sentence-transformers/LaBSE                                              | ✓             | ✓           | ✗          | <b>95.18</b> | <b>94.30</b> | <b>94.06</b> | <b>94.18</b> | <b>94.43</b> |
| English Pivot Translation by M2M 1.2B (Fan et al., 2021) CLSD Evaluation |               |             |            |              |              |              |              |              |
| multilingual-e5-base                                                     | ✓             | ✓           | ✓          | 90.50        | 89.55        | 89.81        | 93.39        | 90.81        |
| multilingual-e5-large                                                    | ✓             | ✓           | ✓          | <b>90.22</b> | 89.95        | <b>91.83</b> | 92.95        | 91.24        |
| gte-multilingual-base                                                    | ✓             | ✓           | ✓          | 89.68        | 89.07        | 90.26        | 92.95        | 90.49        |
| paraphrase-multilingual-mpnet-base                                       | ✓             | ✓           | ✗          | 87.92        | 87.64        | 88.13        | 91.15        | 88.71        |
| sentence-transformers/LaBSE                                              | ✓             | ✓           | ✗          | 90.84        | <b>90.16</b> | 91.83        | <b>94.40</b> | <b>91.81</b> |

### What I've learned today

- Cheap in-domain evaluation is feasible.
- LaBSE handles hard negatives well.
- Skip English pivots when possible.

### Effect of Single-Word Changes

